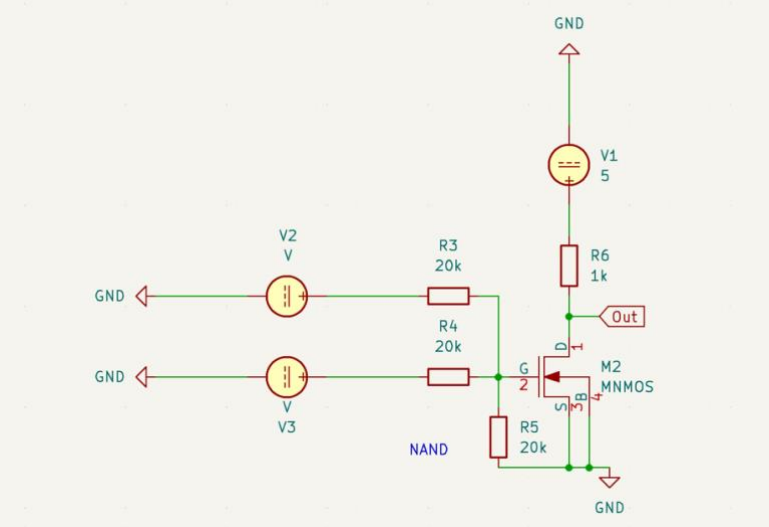
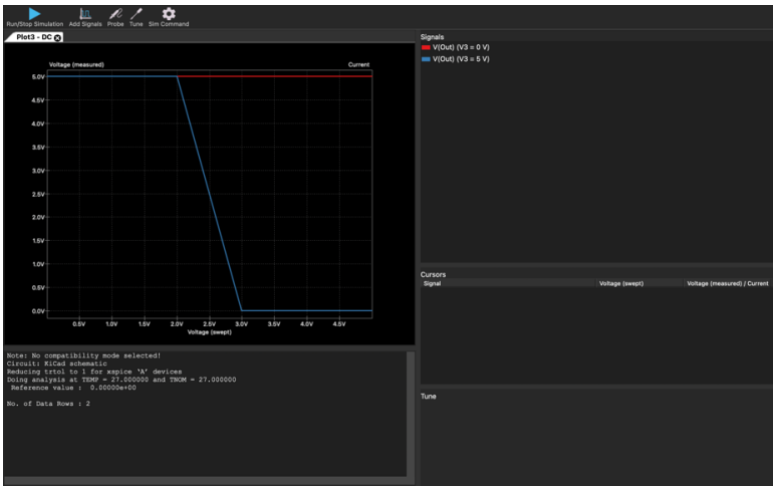
Simulation Assignment 2

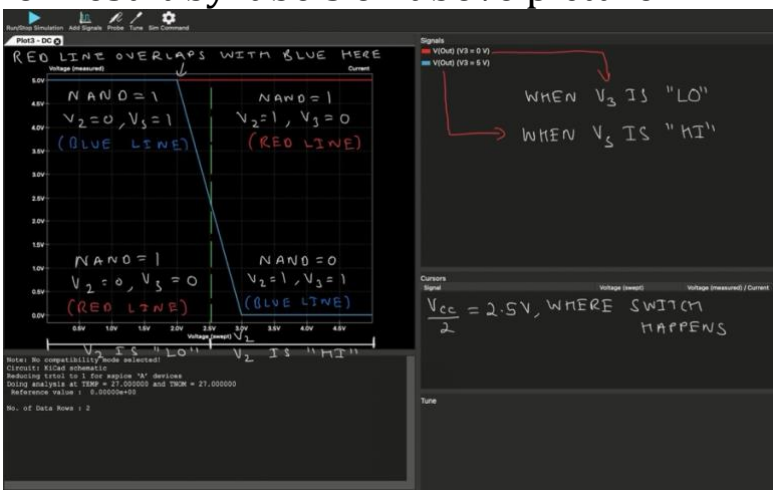
Part A:
Circuit Schematic for NAND Gate



DC Simulation Result:



Explanation of Result by labels on above picture:



Explanation:

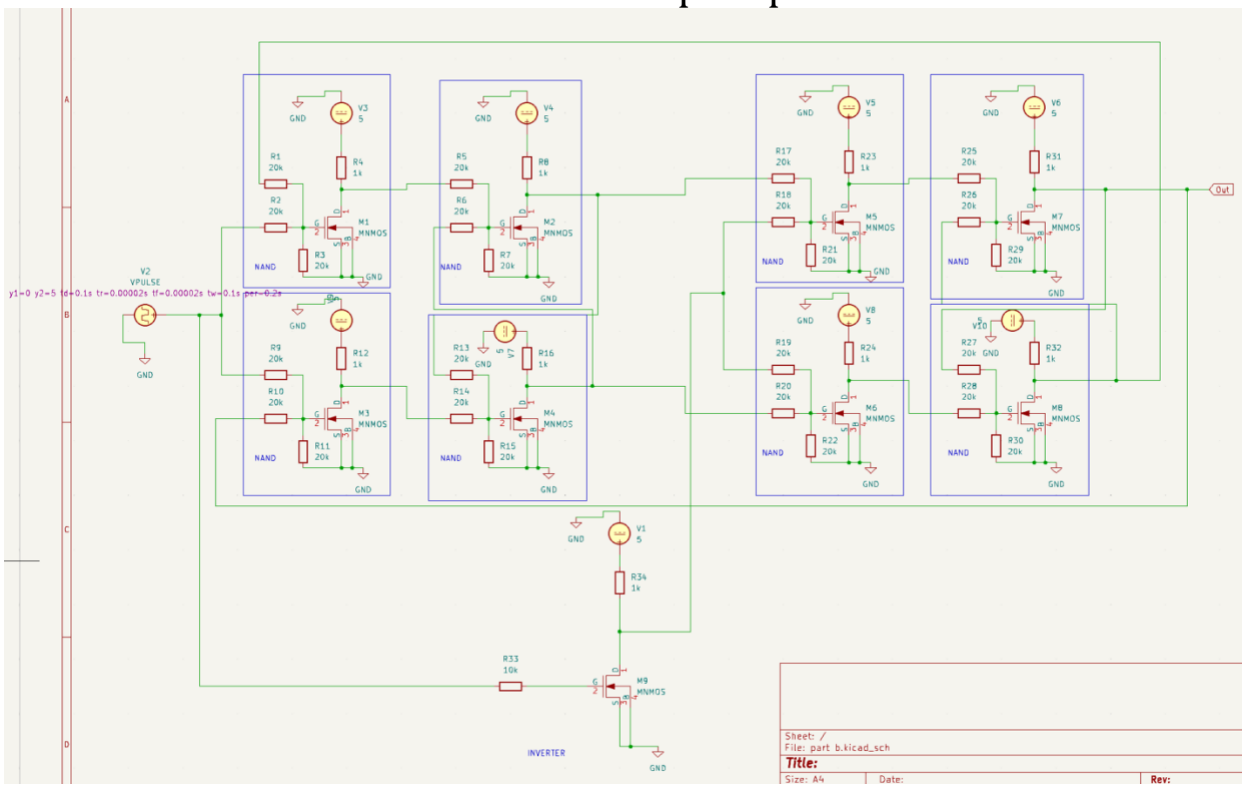
- The blue line represents V_{out} or the output when we fix V_3 as 5V or a 'HI' signal. Then for different values of V_2 (Voltage sweep on V_2) we find out that V_{out} switches from 'HI' to 'LO' exactly at $\frac{V_{cc}}{2} = 2.5V$. When V_2 is between 0-2.5 V the NAND output is correctly a 'HI' signal and when V_2 is between 2.5-5 V the NAND output is correctly a 'LO' signal.
- The red line represents V_{out} or the output when we fix V_3 as 0V or a 'LO' signal. For any value of V_2 we see that V_{out} will always be a 'HI' signal.
- Creating a NAND truth table following the results

V_2/IN_1	V_3/IN_2 (Fixed)	V_{out}/NAND
LO	LO	HI
HI	LO	HI
LO	HI	HI
HI	HI	LO

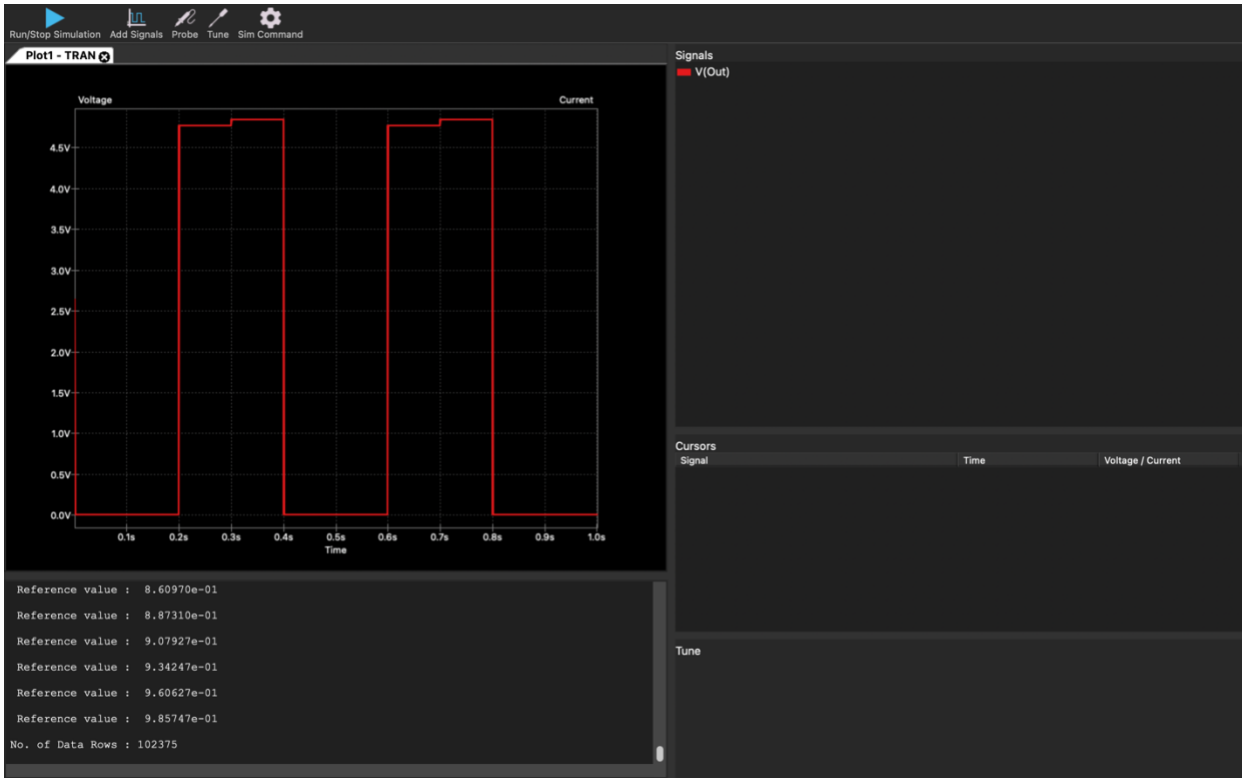
Conclusion: The NAND truth table I created actually matches with the real NAND truth table. So the NAND gate can now be used for part b)

Part B:

Circuit Schematic for Master Slave Flip Flop



Transient Simulation Result



As possible with any circuit, immediately after turn-on the circuit may have some initial variable looking transient effects - but we are now not very interested in those. However, then there should follow a clearly periodic deterministic operation for the master-slave flip-flop when clocked. Can you find it?

After turn-on the output value is around 2.5V but after that there is a clear periodic deterministic operation happening from 0.2s with a period of 0.2s. It looks like a pulse in the result.

How would you describe the operation of the master-slave flip-flop when considering the clock input signal and the flip-flop output signal? Can you also think of where this kind of operation could be required or applied? Write your comments.

When C goes to 1, the slave is isolated; S and R inputs may affect the state of the system. The slave flip-flop is isolated until C goes to 0. When C goes back to 0, information is passed from the master flip-flop to the slave and output is obtained. If S=0 and R=1, the high $\bar{Q}$ output of the master goes to the R input of the slave and the clock forces the slave to reset, thus the slave copies the master. If S=1 and R=0, the high Q output of the master goes to the S input of the slave and the Negative transition of the clock sets the slave, copying the master. If S=1 and R=1, it toggles on the positive transition of the clock and thus the slave toggles on the negative transition of the clock. If S=0 and R=0, the flip flop is disabled and Q remains unchanged.

This could be applied in Sequential logic or storage of bit.

